

§1. 一维问题.

两点边值问题:

$$\begin{cases} -u'' + u = f & \text{in } \Omega = (0, 1) \\ u(0) = u(1) = 0 \end{cases}$$

D.B.C N.B.C

取 $v, v(0) = 0$

$$\int_0^1 f v = \int_0^1 (-u'' + u) v$$

$$= \int_0^1 (u' v' + u v) - \cancel{u' v} \Big|_0^1 \quad \text{边界条件}$$

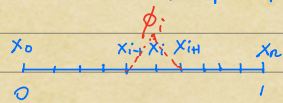
迹定理

(VP) 求 $u \in V = \{v \in H^1 : v(0) = 0\}$ (按 Sobolev $H^1 \subset C^{0, \frac{1}{2}}$)

s.t. $a(u, v) = \int_0^1 (u' v' + u v) dx = (f, v) \quad \forall v \in V$ 自然边界条件为何不须加入到空间中

网格: $0 = x_0 < x_1 < \dots < x_n = 1, x_i$: 节点

$K_i = [x_{i-1}, x_i]$ 单元, $h_i = x_i - x_{i-1}, h = \max_{1 \leq i \leq n} h_i$



有限元空间

逼近空间 $V_h = \{v_h \in C(\Omega) : v_h|_{K_i} \in P_1, 1 \leq i \leq n, v_h(0) = 0\}$.

Finite element

有限元方法 (FEM): 求 $u_h \in V_h$ s.t. $a(u_h, v_h) = (f, v_h) \quad \forall v_h \in V_h$.

节点基: $\phi_i \in V_h, \phi_i(x_j) = \delta_{ij} \quad 1 \leq i \leq n, 0 \leq j \leq n$.

$\forall v_h \in V_h, v_h = v_1 \phi_1 + \dots + v_n \phi_n, v_i = v_h(x_i)$

有限元方程组: $u_h = u_1 \phi_1 + \dots + u_n \phi_n, u_i = u_h(x_i)$

$1 \leq i \leq n$.

$a(u_h, \phi_i) = a(\phi_1, \phi_i) u_1 + \dots + a(\phi_n, \phi_i) u_n = (f, \phi_i) = f_i$

stiffness matrix

$AU = F, A = (a_{ij})_{n \times n}, U = (u_i)_{n \times 1}, F = (f_i)_{n \times 1}$

$a_{ij} = a(\phi_j, \phi_i) = \sum_{m=1}^n \int_{x_{m-1}}^{x_m} (\phi_j' \phi_i' + \phi_j \phi_i) dx$

单元刚度阵

$A^i = (a_{lm}^i)_{2 \times 2}$

$a_{11}^i = \int_{x_{i-1}}^{x_i} \phi_{i-1}' \phi_{i-1}' + \phi_{i-1} \phi_{i-1} = \frac{1}{h_i} + \frac{h_i}{3}$

$a_{22}^i = \int_{x_{i-1}}^{x_i} \phi_i' \phi_i' + \phi_i \phi_i = \frac{1}{h_i} + \frac{h_i}{3}$

$a_{21}^i = a_{12}^i = \int_{x_{i-1}}^{x_i} \phi_{i-1}' \phi_i' + \phi_{i-1} \phi_i = -\frac{1}{h_i} + \frac{h_i}{6}$

单网 $\rightarrow$ 总网.

$a_{ii} = a_{22}^i + a_{11}^{i+1}$

$= \frac{1}{h_i} + \frac{1}{h_{i+1}} + \frac{h_i + h_{i+1}}{3} \quad 1 \leq i \leq n-1$

$a_{nn} = a_{22}^n = \frac{1}{h_n} + \frac{h_n}{3}$

$$a_{i-1,i} = a_{i,i} = -\frac{1}{h_i} + \frac{h_i}{6}$$

$$a_{ij} = 0 \quad |i-j| \geq 2$$

误差估计.

插值. $u_I = I_h u \in V_h$. $u_I(x_i) = u(x_i) \quad 0 \leq i \leq n$.

$$u_I = \sum_{i=1}^n u(x_i) \phi_i$$

$$\hat{e} = \hat{u} - \hat{u}_I$$

定理: (i) $\|u' - u_I'\|_{L^2(K_i)} \leq C h_i \|u''\|_{L^2(K_i)}$

(ii) $\|u - u_I\|_{L^2(K_i)} \leq C h_i^2 \|u''\|_{L^2(K_i)}$

proof: (i) 记 $e = u - u_I$. $x = x_{i-1} + (x_i - x_{i-1}) \hat{x}$

$$\hat{e}(\hat{x}) = e(x)$$

$$\left\| \frac{d\hat{e}}{d\hat{x}} \right\|_{L^2(\hat{K})}^2 = \int_0^1 \left(\frac{d\hat{e}}{d\hat{x}} \right)^2 d\hat{x} = \int_{x_{i-1}}^{x_i} \left(\frac{de}{dx} h_i \right)^2 h_i^{-1} dx$$

$$= h_i \|e'\|_{L^2(K_i)}^2 \leq C h_i^3 \|u''\|_{L^2(K_i)}^2$$

$$= C h_i^3 \|e''\|_{L^2(K_i)}^2$$

$$= C \left\| \frac{d\hat{e}}{d\hat{x}} \right\|_{L^2(\hat{K})}^2$$

只需要证 $\left\| \frac{d\hat{e}}{d\hat{x}} \right\|_{L^2(\hat{K})} \leq C \left\| \frac{d\hat{e}}{d\hat{x}} \right\|_{L^2(\hat{K})}$

$$\hat{e} = \frac{d\hat{e}}{d\hat{x}} \quad \hat{e}(0) = \hat{e}(1) = 0 \quad \int_0^1 \hat{e} d\hat{x} = 0$$

Poincaré 不等式 $\Rightarrow \|\hat{e}\|_{L^2(\hat{K})} \leq C \|\hat{e}'\|_{L^2(\hat{K})}$ 尺度变换

$$a(u, \varphi) = \int_0^1 u' \varphi' + u \varphi dx$$

$$|a(u, \varphi)| \leq \|u\|_{H^1} \|\varphi\|_{H^1}$$

$$a(\varphi, \varphi) = \|\varphi\|_{H^1}^2$$

$$a(u - u_h, \varphi_h) = 0 \quad \forall \varphi_h \in V_h$$

$$\|u - u_h\|_{H^1}^2 \leq a(u - u_h, u - u_h)$$

$$= a(u - u_h, u - u_I)$$

$$\leq \|u - u_h\|_{H^1} \|u - u_I\|_{H^1}$$

$$\|u - u_h\|_{H^1} \leq \|u - u_I\|_{H^1} \leq Ch \|u\|_{H^2}$$

L^2 估计: 对偶问题: $\begin{cases} -w'' + w = u - u_h \\ w(0) = w(1) = 0 \end{cases}$

$$a(w, \varphi) = (u - u_h, \varphi) \quad \forall \varphi \in V$$

$$\|u - u_h\|_{L^2}^2 = (u - u_h, u - u_h) = a(w, u - u_h) = a(u - u_h, w)$$

$$= a(u - u_h, w - w_I)$$

$$\leq \|u - u_h\|_{H^1} \|w - w_I\|_{H^1}$$

$$\leq \|u - u_h\|_{H^1} Ch \|w\|_{H^2}$$

$$\leq \|u - u_h\|_{H^1} Ch \|w\|_{H^2}$$

$$\|w''\|_{L^2}^2 - \int_0^1 w w'' = (u - u_h, w'')$$

$$\|w''\|_{L^2}^2 + \|w'\|_{L^2}^2 \leq \frac{1}{2} \|u - u_h\|_{L^2}^2 + \frac{1}{2} \|w''\|_{L^2}^2$$

§2 Galerkin 方法

抽象框架: V : Hilbert 空间 $a: V \times V \rightarrow \mathbb{R}$, $f \in V'$

(VP) 求 $u \in V$ s.t. $a(u, v) = \langle f, v \rangle \quad \forall v \in V$

$V_h \subset V \quad \dim V_h < \infty$

(GM) 求 $u_h \in V_h$ s.t. $a(u_h, v_h) = \langle f, v_h \rangle \quad \forall v_h \in V_h$

基底: $\{\phi_1, \dots, \phi_N\} \subset V_h \quad u_h = u_1 \phi_1 + \dots + u_N \phi_N$

$a(\phi_i, \phi_j) u_1 + \dots + a(\phi_N, \phi_j) u_N = \langle f, \phi_j \rangle \quad i=1, \dots, N$

$$-u'' + u = f \quad \text{in } \Omega = (0, 1)$$

$$u(0) = u(1) = 0$$

(VP) 求 $u \in V = \{v \in H^1 : v(0) = 0\}$ s.t.

$$a(u, v) = \langle f, v \rangle \quad \forall v \in V$$

(FEM) 求 $u_h \in V_h$ s.t. $a(u_h, v_h) = \langle f, v_h \rangle$

$$\forall v_h \in V_h$$

引理 (Céa) 假设: $\exists \alpha, \beta > 0$ s.t.

$$(i) |a(u, v)| \leq \beta \|u\|_V \|v\|_V \quad \forall u, v \in V$$

$$(ii) a(v, v) \geq \alpha \|v\|_V^2 \quad \forall v \in V$$

$$\text{则} \quad \|u - u_h\|_V \leq \frac{\beta}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V$$

证: $a(u - u_h, v_h) = 0 \quad \forall v_h \in V_h$

$$\alpha \|u - u_h\|_V^2 \leq a(u - u_h, u - u_h)$$

$$= a(u - u_h, u - v_h)$$

$$\leq \beta \|u - u_h\|_V \|u - v_h\|_V \quad \forall v_h \in V_h$$

□

V_h : 多项式 $\rightarrow$ 谱方法

V_h : 分片多项式 $\rightarrow$ 有限元

Petrov-Galerkin 方法: $U_h \subset V \quad V_h \quad \dim U_h = \dim V_h < \infty$

trial

test

(PGM): 求 $u_h \in U_h$ s.t. $a(u_h, v_h) = \langle f, v_h \rangle \quad \forall v_h \in V_h$

引理 (Céa) 设: $\alpha, \beta > 0$ 线性有界算子: $\Pi_h: U_h \rightarrow V_h$

$$\text{s.t. (i)} \quad |a(u, \Pi_h v_h)| \leq \beta \|u\|_V \|\Pi_h v_h\|_V \quad \forall u \in V, v_h \in U_h$$

$$(ii) \quad a(\Pi_h v_h, \Pi_h v_h) \geq \alpha \|\Pi_h v_h\|_V^2 \quad \forall v_h \in U_h$$

$$\text{则} \quad \|u - u_h\|_V \leq \left(1 + \frac{\beta}{\alpha}\right) \inf_{v_h \in U_h} \|u - v_h\|_V$$

证: $a(u - u_h, \Pi_h v_h) = 0 \quad \forall v_h \in U_h$

$$\alpha \|u_h - \Pi_h v_h\|_V^2 \leq a(u_h - \Pi_h v_h, u_h - \Pi_h v_h)$$

$$= a(u - \Pi_h v_h, \Pi_h(u_h - v_h))$$

$$\leq \beta \|u - \Pi_h v_h\|_V \|u_h - \Pi_h v_h\|_V$$

$$\|u_h - \Pi_h v_h\|_V \leq \frac{\beta}{\alpha} \|u - \Pi_h v_h\|_V$$

$$\|u - u_h\|_V \leq \|u - \Pi_h v_h\|_V + \|\Pi_h v_h - u_h\|_V$$

□

非标准 GM: $V_h \not\subset V$, $\dim V_h < \infty$

(VP) 求 $u \in V$ s.t. $a(u, v) = (f, v) \quad \forall v \in V$

$a_h: (V+V_h) \times (V+V_h) \rightarrow \mathbb{R}$ 双线性

$f_h \in V_h$

(NGM): 求 $u_h \in V_h$ s.t. $a_h(u_h, v_h) = (f_h, v_h) \quad \forall v_h \in V_h$

引理: (strong) 设 $\|\cdot\|_h$ 是 $V+V_h$ 上的范数, $\exists \alpha, \beta > 0$ s.t.

(i) $|a_h(u, v)| \leq \beta \|u\|_h \|v\|_h \quad \forall u, v \in V+V_h$

(ii) $a_h(v_h, v_h) \geq \alpha \|v_h\|_h^2 \quad \forall v_h \in V_h$

则 $\|u - u_h\|_h \leq (1 + \frac{\beta}{\alpha}) \inf_{v_h \in V_h} \|u - v_h\|_h + \frac{1}{\alpha} \sup_{0 \neq w_h \in V_h} \frac{|(f_h, w_h) - a_h(u, w_h)|}{\|w_h\|_h}$

证: $\alpha \|u_h - v_h\|_h^2 \leq a_h(u_h - v_h, u_h - v_h)$

$= a_h(u - v_h, u_h - v_h) + (f_h, u_h - v_h) - a_h(u, u_h - v_h)$

$\leq \beta \|u - v_h\|_h \|u_h - v_h\|_h + \|u_h - v_h\|_h \sup_{0 \neq w_h \in V_h} \frac{|(f_h, w_h) - a_h(u, w_h)|}{\|w_h\|_h}$

相容性误差

应用: (P) $\begin{cases} -\nabla \cdot (a \nabla u) + cu = f & \text{in } \Omega \\ u|_{\partial} = 0 & \text{或 } a \nabla u \cdot n = g & \text{或 } a \nabla u \cdot n + qu = g, \text{ on } \partial. \end{cases}$

(VP) 求 $u \in V$ s.t. $a(u, v) = (f, v) \quad \forall v \in V$

(GM) 求 $u_h \in V_h$ s.t. $a(u_h, v_h) = (f, v_h) \quad \forall v_h \in V_h \subset V$

非齐次 Dirichlet B.C. $u|_{\partial} = g$

(VP) 求 $u \in H^1$, $u|_{\partial} = g$ s.t. $a(u, v) = (f, v) \quad \forall v \in H_0^1$

(GM) 求 $u_h \in V_h$, $u_h|_{\partial} = g_h$ s.t. $a(u_h, v_h) = (f, v_h) \quad \forall v_h \in V_h \cap H_0^1$

非线性空间不是 PGM

引理: (Cea) ...

则 $\|u - u_h\|_{H^1} \leq (1 + \frac{\beta}{\alpha}) \inf_{\substack{v_h \in V_h \\ v_h|_{\partial} = g_h}} \|u - v_h\|_{H^1}$

§3 有限元空间

有限元: 三元组 (K, P, N)

(i) K : 边界分片光滑闭区域 $\leftarrow$ 单元

(ii) P : K 上有限维函数空间 $\leftarrow$ 形函数, $\dim P = n$

(iii) $N = \{N_1, \dots, N_n\}$ 是 P' 的一组基 $\leftarrow$ 自由度

引理: (iii) $\Leftrightarrow$ (iii)': 若 $v \in P$, $N_i(v) = 0 \quad 1 \leq i \leq n$ 则 $v \equiv 0$

证: $\{\psi_1, \dots, \psi_n\}$ 是 P 的基 $\forall v \in P, \exists x_1, \dots, x_n$ s.t. $v = x_1 \psi_1 + \dots + x_n \psi_n$

(iii)' $\Leftrightarrow N_i(\psi_j) x_1 + \dots + N_i(\psi_n) x_n = 0 \quad 1 \leq i \leq n$ 只有零解

$\Leftrightarrow M = (N_i(\psi_j))_{n \times n}$ 可逆

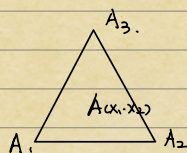
(iii) $\Leftrightarrow \forall L \in P' \exists y_1, \dots, y_n$ s.t. $L = y_1 N_1 + \dots + y_n N_n$

$\Leftrightarrow y_1 N_1(\psi_i) + \dots + y_n N_n(\psi_i) = L(\psi_i) \quad 1 \leq i \leq n$

$\Leftrightarrow M^T$ 可逆.

□

三角形上的有限元
Lagrange. 线性元



$$K = \triangle A_1 A_2 A_3 \quad P = P_1$$

$$N_i(\psi) = \psi(A_i)$$

节点基 $\{\lambda_1, \lambda_2, \lambda_3\}$

$$N_i(\psi) = 0 \quad 1 \leq i \leq n \quad \text{则} \quad \psi \equiv 0$$

$$\lambda_1 + \lambda_2 + \lambda_3 = 1$$

重心坐标: 在笛卡尔坐标系下点 $X = (x_1, x_2)$ 的重心坐标定义为 $(\lambda_1(x), \lambda_2(x), \lambda_3(x))$ $\alpha_i = (a_i, b_i)$ 是 A_i 坐标,
$$X = \alpha_1 \lambda_1 + \alpha_2 \lambda_2 + \alpha_3 \lambda_3$$

$$S = S_{\triangle A_1 A_2 A_3} = \frac{1}{2} \begin{vmatrix} a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \\ a_3 & b_3 & 1 \end{vmatrix}$$

$$S_{\triangle A_1 A_2 A_3} = \frac{1}{2} \begin{vmatrix} x & y & 1 \\ a_2 & b_2 & 1 \\ a_3 & b_3 & 1 \end{vmatrix}$$

有限元例子见教材

有限元空间 H^m .

引理 $M_h = \{K_1, \dots, K_J\}$ 是 $\mathbb{R}^2$ 的一个剖分

$$(i) \quad \overline{\Omega} = \bigcup_{K \in M_h} K \quad (ii) \quad K_i \cap K_j = \emptyset \quad i \neq j$$

$$m \geq 1. \quad \psi|_{K_i} \in C^m(K_i) \quad \text{则} \quad \psi \in H^m(\Omega) \Leftrightarrow \psi \in C^{m-1}(\Omega)$$

证: " $\Leftarrow$ " 设 $m=1$ $\psi \in L^2 \quad \nabla \psi|_{K_i} = \nabla(\psi|_{K_i}) \quad 1 \leq i \leq J.$

$$\forall \varphi \in (C_0^\infty(\Omega))^d$$

$$\int_{\Omega} w \cdot \varphi = \sum_{i=1}^J \int_{K_i} \nabla \psi \cdot \varphi = \sum_{i=1}^J \left(- \int_{K_i} \psi \nabla \cdot \varphi + \int_{\partial K_i} \psi \varphi \cdot n_{K_i} \right)$$

$$= - \int_{\Omega} \psi \nabla \cdot \varphi + \sum_{i=1}^J \int_{\partial K_i} \psi \varphi \cdot n_{K_i}$$

$$\Rightarrow \nabla \psi = w \in L^2$$

$$\Rightarrow \psi \in H^1$$

Cell 元

C^1 元 = 继 最小自由度 18 可以去详细了解.

§4 有限元方程组的形成

$$(P) \quad \begin{cases} -\nabla \cdot (a \nabla u) + cu = f & \text{in } \Omega \\ a \nabla u \cdot n|_{\Gamma} = 0 \end{cases}$$

$$(VP) \quad \text{求 } u \in H^1 \quad \text{s.t.} \quad a(u, \varphi) = (f, \varphi) \quad \forall \varphi \in H^1 \\ a(u, \varphi) = \int_{\Omega} (a \nabla u \cdot \nabla \varphi + cu \varphi)$$

$$M_h: \text{三角剖分} \quad \{x_j: 1 \leq j \leq J\}$$

$$V_h = \{\varphi_h \in C(\bar{\Omega}): \varphi_h|_K \in P_1 \quad \forall K \in M_h\}.$$

$$(FEM) \quad \text{求 } u_h \in V_h \quad \text{s.t.} \quad a(u_h, \varphi_h) = (f, \varphi_h) \quad \forall \varphi_h \in V_h$$

$$\text{节点基: } \{\phi_j: 1 \leq j \leq J\}, \quad \phi_j(x_i) = \delta_{ij}$$

$$u_h = \sum u_j \phi_j, \quad u_j = u_h(x_j)$$

$$\sum_j a(\phi_j, \phi_i) u_j = (f, \phi_i) \quad 1 \leq i \leq J$$

$$A = (a_{ij})_{J \times J} \quad a_{ij} = \int_{\Omega} (a \nabla \phi_j \cdot \nabla \phi_i + c \phi_j \phi_i) \quad \text{刚度阵.}$$

$$a_{ij} = \sum_{K \in M_h} \int_K (a \nabla \phi_j \cdot \nabla \phi_i + c \phi_j \phi_i)$$

$$\text{单元刚度阵: } A^K = (a_{\ell m}^K)_{3 \times 3}$$

$$a_{\ell m}^K = \int_K (a \nabla \lambda_m \nabla \lambda_{\ell} + c \lambda_m \lambda_{\ell})$$

K_p : K 的第 p 个顶点的整体编号.

$$\phi_{K_p}|_K = \lambda_p$$

$$a_{ij} = \sum_{\substack{K \in M_h \\ b.m.}} a_{\ell m}^K$$

$$K_{\ell} = i \quad K_m = j$$

数值积分

代数精度 p .